From Static to Contextual: Distributional Semantics and What's Inside BERT

NLP CS60075 Assignment II – Project 5, dataset: UD English EWT

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Abstract

We build classical PPMI+SVD word vectors from the UD English EWT training text (Task 1), reproduce the Hewitt & Manning (2019) structural probe on frozen `bert-base-uncased` (Task 2), and turn the most syntactic BERT layer into static word vectors that we compare with Task 1 on semantic and structural tests (Task 3). The probe reproduces the paper almost exactly: 79.5% UUAS at layer 8 on EWT test vs. 79.8% on PTB, with the same mid-layer peak (layers 6–8 are statistically tied). BERT-derived static vectors are far better than PPMI-SVD on meaning (WordSim-353 ρ 0.577 vs. 0.104; analogy 67.7% vs. 0.51%), but *worse* on the tree-recovery check (17.9% vs. 24.8% UUAS on 660 test sentences, close to a random tree). The two evaluations therefore disagree. We show with measured diagnostics why: syntax lives in the context that pooling averages away, and BERT-static similarity is *paradigmatic* (substitutes) while dependency edges need *syntagmatic* association (partners), which the count-based vectors partly have.

1 Introduction

The distributional hypothesis (Harris, 1954) says a word's meaning can be read from the contexts it occurs in. Count-based models turn this into one fixed vector per word (Levy and Goldberg, 2014; Levy et al., 2015); BERT (Devlin et al., 2019) gives every *token* its own, sentence-dependent vector. On one treebank, UD English EWT (Silveira et al., 2014; de Marneffe et al., 2021), the project asks:

- **T1:** how good are classical PPMI+SVD vectors trained on 200k words, and do they contain any tree structure?
- **T2:** can a linear structural probe recover dependency trees from BERT, and which layer ℓ^* encodes them best?
- **T3:** if layer-ℓ^* vectors are averaged into static vectors, are they better than T1 on word meaning and on tree structure – and do the two answers agree?

Main findings

1. **T1 is limited by data:** WS-353 $\rho = 0.104$ overall but 0.317 on frequent words; its MST beats a random tree but not the linear chain.
2. **T2 reproduces the paper:** 79.5% UUAS and NSpr 0.859 at ℓ^*=8, within 0.3 pt UUAS of the paper; context adds ≈21 pt over the non-contextual layer 0, mostly for function words.
3. **T3 wins semantics but loses the structural check:** the two evaluations disagree, and §5.1 tests each explanation quantitatively.

All numbers are generated by a script from the experiment outputs; seeds are fixed (§8).

2 Data, benchmarks and shared protocol

Treebank

- UD English EWT (web text: reviews, e-mails, newsgroups, Q&A, blogs), loaded from the Hugging Face `universal-dependencies/universal_dependencies` repository at the immutable tag `ud2.18-loader-v2.3.0`.
- Syntactic words are used (multi-word-token surface forms and empty nodes excluded); trees come from `head`, punctuation from UPOS.
- All trees are valid; only 10 punctuation tokens in the whole treebank are non-leaves.

split	sentences	words	mean len.	max len.	PUNCT tokens	sents. with MWT
train	12,544	204,578	16.3	159	23,596	2,192
dev	2,001	25,148	12.6	75	3,075	303
test	2,077	25,094	12.1	81	3,096	302

Table 1: UD English EWT splits. Train has 16,654 lowercased word types, 47% of them singletons; 7.5% of test tokens are unseen in train.

Benchmarks

- WordSim-353 (353 pairs, Finkelstein et al. 2002) and the Google analogy set (19,544 questions, 14 sections, Mikolov et al. 2013): the copies shipped with `gensim` (pinned commit, SHA-256 recorded), lowercased.
- **Coverage is the central practical issue.** Only 248/353 WS pairs (70.3%) and 5,727/19,544 analogy questions (29.3%) have all their words in the vocabulary; a covered WS word occurs only 11 times (median).
- T1 and T3 use *the same vocabulary* (all lowercased train types), so every comparison is on the same items.
- Two numbers per benchmark:
  - *all items* – the spec-literal number: all 353 pairs, OOV pairs get the median model similarity (a neutral rank); OOV analogy questions count as wrong;
  - *covered items* – the informative comparison.

Structural check (T1 and T3)

- **Sample:** 20 test sentences drawn once (seed 42) from the 660 test sentences with 5–40 non-punctuation words, all in the train vocabulary (a static model has no vector for an OOV word); the same 20 IDs are reused in T3.
- **Tree:** Prim's MST over pairwise cosine distances of the words' vectors.
- **Score:** undirected attachment (UUAS) against the gold tree, punctuation ignored (as in the probe paper).
- **Extras**, because 20 sentences (234 edges) are few:
  - the whole eligible pool of 660 sentences;
  - bootstrap 95% CIs (1000 resamples of sentences) and paired tests between systems;
  - two baselines: the *linear chain* (attach each word to the next; the paper's LINEAR) and the expected UUAS of a *uniformly random tree*.

3 Task 1 – PPMI + SVD baseline

Method

- **Counts:** word–word co-occurrence over the train sentences, symmetric window of 5 words on each side, within the sentence, unweighted.
- **Weighting:** PPMI = $\max(0, \log \frac{P(w,c)}{P(w)P(c)})$ (Levy and Goldberg, 2014).
- **Reduction:** truncated SVD to 100 dimensions (randomized solver, seed 42); word vectors $U\Sigma$.
- **Evaluation:** cosine similarity; analogies by 3CosAdd, $\arg\max_x \cos(x, b - a + c)$ with the three query words excluded.
- **Fair-baseline checks (beyond the spec):** the Levy et al. (2015) hyper-parameters (context-distribution smoothing $\alpha = 0.75$, eigenvalue weight $p \in \{0, 0.5\}$) and all-but-the-top post-processing.

Results (Table 2)

- **WS-353:** $\rho = 0.048$ over all 353 pairs; 0.104 [−0.03, 0.22] on the covered ones.
- **Analogies:** 0.15% (all) / 0.51% (covered; 29 of 5,727 correct).
- **Structural check:** UUAS 20.5% [16.3, 26.2] on the 20 sentences – +5.3 pt above a random tree (paired CI [1.4, 9.2], significant), but −22.6 pt below the linear chain (43.2%).

	WordSim-353 ρ			Analogy acc. (%)		MST UUAS (%)	
PPMI-SVD variant	covered (248) [95% CI]	all 353	freq. ≥20	covered	all	20 sent.	pool (660)
spec: α=1, p=1 (UΣ)	0.104 [−0.03, 0.22]	0.048	0.317	0.51	0.15	20.5	24.8
α=0.75, p=1	0.060 [−0.07, 0.20]	0.043	0.199	0.65	0.19	15.8	20.8
α=1, p=0.5	0.129 [−0.00, 0.24]	0.091	0.289	0.37	0.11	20.9	24.6
α=1, p=0 (U)	0.134 [0.00, 0.25]	0.089	0.320	0.21	0.06	22.6	24.5
α=0.75, p=0.5	0.097 [−0.04, 0.22]	0.066	0.174	0.42	0.12	16.2	20.8
spec + all-but-the-top (D=1)	0.153	0.103	0.294	0.42	–	20.5	24.5
linear chain (attach to next word)	–	–	–	–	–	43.2	43.0
random tree (expected)	–	–	–	–	–	15.2	17.1

Table 2: Task 1 results. "covered" = items whose words are all in the vocabulary; "all" = all items (spec-literal). "freq. ≥20" = the 32 covered pairs whose two words occur ≥20 times in train. Pool = all 660 eligible test sentences.

Why the numbers are low: data, not implementation

1. **Frequent words are fine:** ρ rises from 0.104 (all covered pairs) to 0.248 (both words ≥5 occurrences), 0.283 (≥10) and 0.317 (≥20).
2. **Neighbours of frequent words are sensible** co-occurrence partners (Table 9).
3. **No hyper-parameter rescues it:** the best variant ($p = 0$) reaches $\rho = 0.134$; no setting lifts analogies above 1%.
4. **Analogies need four well-estimated vectors:** the in-vocabulary words of the capital, family, comparative and currency sections have median train counts of only 6–11.

Which tree edges PPMI-SVD finds (recall on the pool)

- **Content words that co-occur:** amod 0.40, advmod 0.36, conj 0.34, obj 0.31.
- **Almost no function-word attachments:** cc 0.05, case 0.12, mark 0.13, det 0.14 – function words co-occur with everything, so their vectors are uninformative.

4 Task 2 – Reproducing the structural probe

Features

- Each EWT sentence is fed to `google-bert/bert-base-uncased` (revision `86b5e09`) as pre-split words.
- All 13 hidden states are kept: layer 0 = embedding layer, layers 1–12 = transformer blocks (HF `hidden_states[`ℓ`]`).
- A word's vector is the mean of its word-piece vectors, as in the paper.
- No sentence exceeds 510 word pieces (max. 335), so nothing is truncated; features are cached once (float16, 5.1 GB, 410 s on one L40S).

Probes and training

- **Distance probe:** $d_B(h_i, h_j) = \|B(h_i - h_j)\|_2^2$, trained to match the gold tree distance.
- **Depth probe:** $\|Bh_i\|_2^2$, trained to match the gold depth.
- $B \in \mathbb{R}^{k\times 768}$, $k \in \{32, 64, 128\}$, initialised $\mathcal{U}(-0.05, 0.05)$.
- **Loss (the paper's):** L1 between prediction and gold, summed over word pairs (words), divided by n^2 (n) per sentence, averaged over the batch; gold distances and depths come from `head` by BFS over all tokens.
- **Optimisation:** Adam, lr 10^{-3}, batches of 20 sentences, 30 epochs, train split; 13 layers × 3 ranks × 2 probes = 78 probes (layer 0 added as a non-contextual baseline).
- **Schedule (paper, App. A):** when the dev loss does not improve, lr ×0.1 (optimizer reset); the best-dev checkpoint is kept. §4.2 shows why this matters.

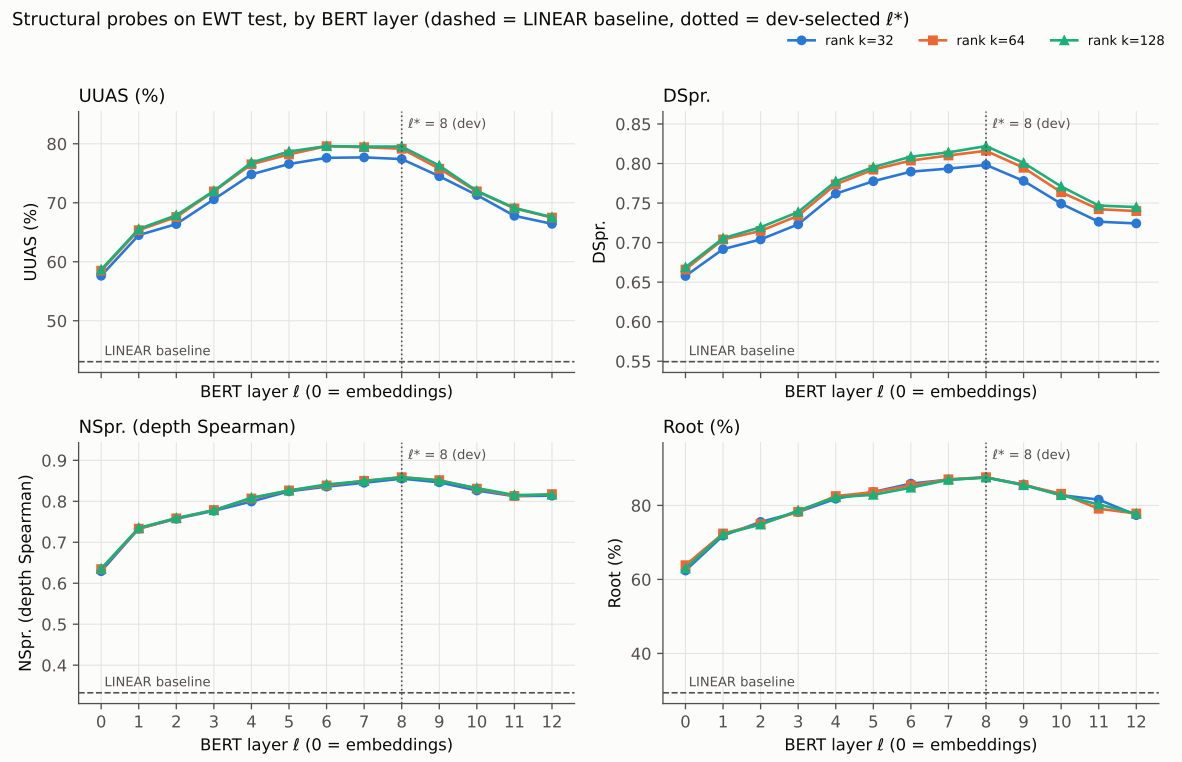


Figure 1: Probe metrics on EWT test vs. BERT layer, one line per rank (dashed = LINEAR baseline, dotted = dev-selected ℓ^*).

Evaluation (as in the official `reporter.py`)

- **UUAS:** MST of predicted distances vs. the gold tree over non-punctuation words, micro-averaged over all test sentences.
- **DSpr:** per-word Spearman of predicted vs. gold distance rows. **NSpr** ("depth Spearman"): per-sentence Spearman of predicted vs. gold depth. Both are averaged per sentence length and macro-averaged over lengths 5–50.
- **Root%:** the shallowest non-punctuation word is the gold root.
- ℓ^* is chosen on **dev** UUAS (best rank per layer); test is never used for a choice.

Results (Fig. 1, Table 3)

- **Layer curve:** UUAS rises from 58.7% (layer 0) to a plateau over layers 6–8 and falls to 67.6% at layer 12; the depth probe has the same shape.
- **Best layer:** dev selects ℓ^*=8 (dev UUAS 80.0%); depth also peaks at 8 on dev and test.
- **Significance:** on test, layers 6–8 are statistically tied with 8 (the test argmax, layer 6, wins by 0.08 pt); every other layer is significantly worse.
- **Rank:** at ℓ^*, UUAS 77.4 / 79.2 / 79.5% and NSpr 0.855 / 0.859 / 0.859 for $k = 32/64/128$. The gain stops around $k = 64$, as in the paper (Fig. 5): the tree lives in a ≈64-dimensional subspace.
- **Convergence:** every probe reached its best dev loss by epoch 8–17.

What context adds

- **Layer 0 is already strong:** word-piece + position embeddings with no context reach 58.7% – far above LINEAR (43.1%) and close to the paper's non-contextual PROJ0 (59.8). Word identity plus position explains much of a tree.
- **Context adds ≈21 pt,** the evidence that BERT's context encodes syntax.
- **Biggest gains for function words,** which have no lexical cue to their head (recall, layer 0 → 8): case 0.35→0.73, cc 0.40→0.82, mark 0.56→0.80, obl 0.42→0.69.
- **Smallest gains for adjacent pairs:** compound 0.82→0.87, amod 0.76→0.87.
- **At ℓ^*:** easiest obj (0.93), det (0.91), aux (0.89); hardest conj (0.56) – UD attaches every conjunct to the first,

	UUAS (%), rank k				ΔUUAS vs. ℓ^* (pt)	NSpr (depth Spearman), k			
ℓ	32	64	128	DSpr	paired 95% CI (k=128)	32	64	128	Root%
0	57.6	58.5	58.7	0.669	−20.9 [−21.6, −20.1]	0.629	0.634	0.636	63.0
1	64.5	65.4	65.6	0.705	−14.0 [−14.7, −13.2]	0.733	0.733	0.735	72.2
2	66.4	67.6	67.9	0.719	−11.6 [−12.4, −10.9]	0.757	0.758	0.759	74.8
3	70.6	71.9	72.0	0.739	−7.5 [−8.2, −6.8]	0.777	0.779	0.778	78.3
4	74.8	76.5	76.8	0.777	−2.7 [−3.3, −2.1]	0.799	0.807	0.809	82.2
5	76.6	78.2	78.7	0.795	−0.8 [−1.4, −0.3]	0.824	0.827	0.826	82.9
6	77.6	79.6	79.6	0.809	+0.1 [−0.4, 0.6]†	0.836	0.839	0.841	84.8
7	77.7	79.4	79.5	0.814	0.0 [−0.4, 0.5]†	0.845	0.850	0.850	87.0
8	77.4	79.2	79.5	0.822	–	0.855	0.859	0.859	87.5
9	74.5	75.8	76.3	0.801	−3.2 [−3.7, −2.8]	0.846	0.852	0.851	85.5
10	71.3	71.9	72.0	0.771	−7.5 [−8.1, −7.0]	0.826	0.831	0.832	82.8
11	67.8	69.1	69.1	0.742	−10.4 [−11.0, −9.8]	0.812	0.813	0.815	80.3
12	66.4	67.5	67.6	0.745	−12.0 [−12.5, −11.4]	0.814	0.818	0.816	77.8
LINEAR		43.1		0.549	–		0.333		29.4

Table 3: Task 2 on EWT test, all layers and ranks. DSpr/Root% use the dev-best rank of the layer. The Δ column is a paired bootstrap of each layer vs. ℓ^* over the same sentences; † = 95% CI contains 0 (not significantly different). Shaded: ℓ^*.

creating long arcs – and obl (0.69), the PP-attachment ambiguity (heatmap in App. B).

4.1 Comparison with the paper

	system	UUAS	DSpr	Root%	NSpr
Paper (PTB WSJ)	LINEAR	48.9	0.58	2.9	0.27
Paper (PTB WSJ)	PROJ0	59.8	0.73	64.4	0.75
Paper (PTB WSJ)	ELMO1	77.0	0.83	86.5	0.87
Paper (PTB WSJ)	BERTBASE7	79.8	0.85	88.0	0.87
Paper (PTB WSJ)	BERTLARGE16	81.7	0.87	90.1	0.89
Ours (EWT)	LINEAR (word order only)	43.1	0.549	29.4	0.333
Ours (EWT)	BERT layer 0 (embeddings + position)	58.7	0.669	63.0	0.636
Ours (EWT)	**BERT-base-uncased layer 8** (paper schedule)	79.5	0.822	87.5	0.859
Ours (EWT)	BERT layer 8, spec-literal (constant lr, last epoch)	62.0	0.760	58.6	0.803

Table 4: Our numbers vs. Hewitt and Manning (2019) Table 1 (PTB WSJ test, BERT-base *cased*, Stanford Dependencies).

- **The result reproduces:** within 0.3 pt UUAS, 0.03 DSpr, 0.5 pt Root% and 0.01 NSpr of the paper's BERTBASE7.
- **Non-contextual baseline:** our layer-0 probe matches their PROJ0 (UUAS 58.7 vs. 59.8).
- **Layer numbering:** the paper counts BERT-base layers 0–11 without the embedding layer, so its "layer 7" is our HF layer 8 (or 7 under the other reading); layers 6–8 are tied, so the comparison holds either way.
- **Likely causes of the small DSpr/NSpr gap** (plausible, not isolated):
  1. **data scale** – 12.5k train sentences vs. ≈40k in PTB;
  2. **genre** – web text with typos and fragments vs. edited news;
  3. **annotation** – UD's content-head trees (prepositions and copulas are dependents) vs. Stanford Dependencies;
  4. **casing** – uncased vs. cased model;
  5. **budget** – rank ≤128 and 30 epochs vs. full rank and up to 40 epochs.
- **LINEAR differs more** (UUAS 43.1 vs. 48.9; Root% 29.4 vs. 2.9): EWT sentences are short and often *start with their root* ("Say . . . ", "Thanks . . . ", fragments), whereas WSJ sentences start with a subject.

	WordSim-353 ρ		Analogy acc. (%)			MST UUAS (%)	
system	covered [95% CI]	all 353	covered	sem / syn	all	20 sent. [95% CI]	pool
T1 PPMI-SVD (spec)	0.104 [−0.03, 0.22]	0.048	0.5	2.1 / 0.3	0.1	20.5 [16.3, 26.2]	24.8
T1 PPMI-SVD, best variant (p=0)	0.134 [0.00, 0.25]	0.089	0.2	0.9 / 0.1	0.1	22.6 [17.8, 28.4]	24.5
T3 BERT-static, layer 8	0.577 [0.47, 0.66]	0.452	67.7	74.5 / 66.6	19.8	18.4 [13.6, 26.0]	17.9
T3 + all-but-the-top (D=7)	0.682 [0.61, 0.74]	0.539	71.0	79.0 / 69.7	20.8	19.7 [14.9, 26.2]	19.6
T2 probe, layer 8 (contextual, supervised; reference)	–	–	–	–	–	81.2 [75.9, 85.9]	82.9
linear chain / random tree	–	–	–	–	–	43.2 / 15.2	43.0 / 17.1

Table 5: Task 3 vs. Task 1 (identical items), with the Task 2 probe as a reference on the same 20 sentences and pool.

	WordSim-353	Analogy (McNemar)		MST ΔUUAS (pt) [95% CI]	
A − B	$\Delta\rho$ [95% CI]	only A / only B	p	20 sentences	pool (660)
T3 − T1	+0.474 [0.36, 0.61]	3,848 / 0	$< 10^{-300}$	−2.1 [−6.4, 3.2]	−6.9 [−8.0, −5.8]
T3+ABTT − T1	+0.579 [0.46, 0.71]	4,036 / 0	$< 10^{-300}$	−0.9 [−6.6, 5.6]	−5.1 [−6.2, −4.1]
T3+ABTT − T3	+0.105 [0.05, 0.16]	276 / 88	1×10^{-23}	+1.3 [−3.2, 5.6]	+1.7 [1.0, 2.6]
T2 probe − T3	–	–	–	+62.8 [53.2, 69.6]	+65.0 [63.8, 66.3]

Table 6: Paired tests on identical items. Analogy: number of questions solved by only A / only B, exact McNemar p.

4.2 Control run: the assignment's literal training schedule

- **Why:** the assignment specifies Adam with lr 10^{-3} for 30 epochs; we added the paper's lr decay and best-dev selection.
- **Run:** every layer retrained ($k = 128$) with constant lr 10^{-3}, keeping the epoch-30 probe.
- **Same conclusion:** ℓ^* is unchanged (8 on dev and test) and the curve keeps its rise–peak–fall shape.
- **Much lower level:** UUAS at ℓ^* drops to 62.0% (DSpr 0.760, NSpr 0.803, Root 58.6%; Table 4).
- **Cause – non-convergence:** at layer 8 the *training* loss stays at 0.62 (vs. 0.42 with decay) and the dev loss oscillates between 0.54 and 0.59 for 30 epochs. With an L1 loss the gradient does not shrink near the optimum, so Adam's steps stay ≈lr and circle the minimum; the decay lets the probe converge.
- **Verdict:** the deviation is necessary for a faithful reproduction, and the main conclusion (ℓ^*=8) does not depend on it.

5 Task 3 – BERT-derived static vectors

Method

- **Pooling:** for each of the 16,654 T1 word types, average its layer-ℓ^* vectors (each the mean of its word pieces, computed *in context*) over all its occurrences in EWT train – the mean/mean pooling that Bommasani et al. (2020) found best.
- **Identical protocol:** same vocabulary and row order as T1; WS-353, analogy and the 20-sentence MST run through *the same function* as T1.

Additions beyond the spec

(a) **All-but-the-top** (Mu and Viswanath, 2018): subtract the mean and remove the top $D = d/100$ principal directions ($D = 7$ for BERT), because pooled BERT vectors are anisotropic (Ethayarajh, 2019); also applied to T1 ($D = 1$) so post-processing is not credited to BERT.

(b) **Paired significance tests** on identical items: bootstrap of $\Delta\rho$ over pairs, exact McNemar over analogy questions, bootstrap of ΔUUAS over sentences.

(c) **Layer sweep** over layers 0–12.

(d) **Diagnostic ladder** of MSTs on the same sentences.

(e) **Combination variants:** concatenated L2-normalised [T1; T3] vectors, and T3 vectors seen through the T2 probe matrix B.

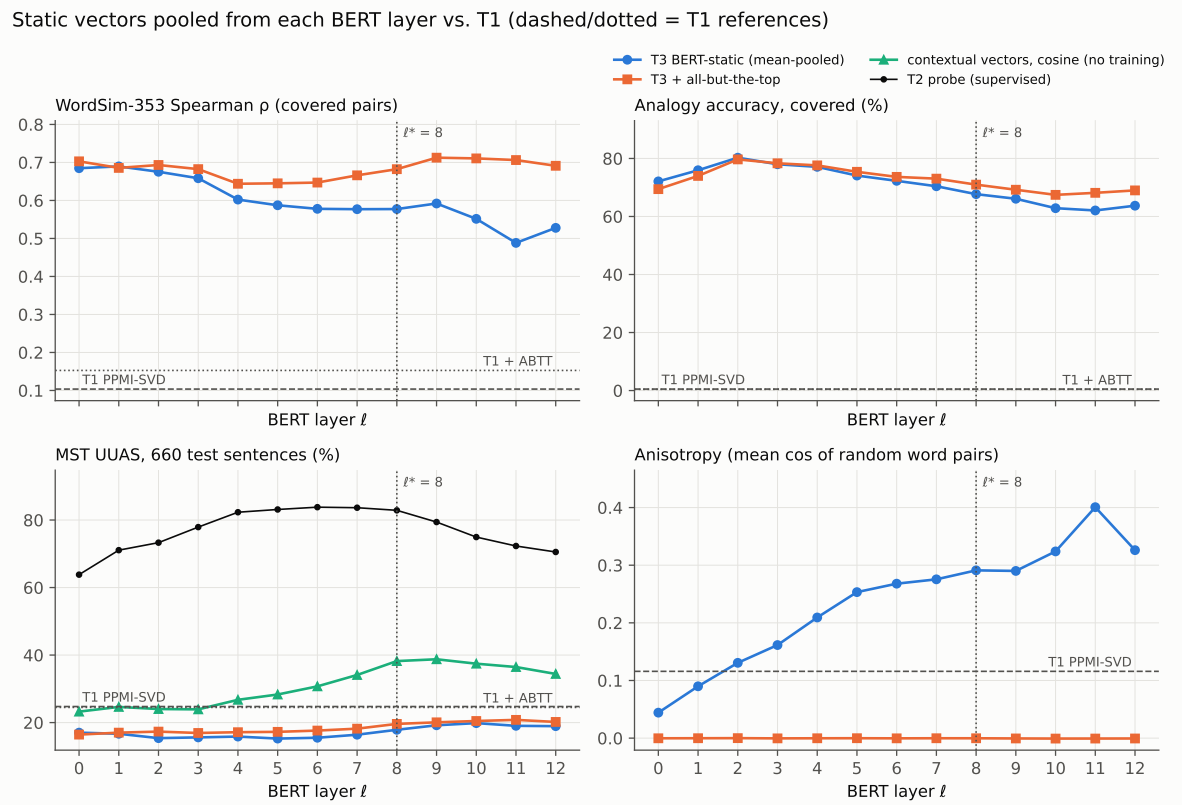


Figure 2: Static vectors pooled from every BERT layer (raw and with all-but-the-top) vs. T1 references; the MST panel also shows raw contextual vectors (cosine, no training) and the T2 probe on the same 660 sentences. Numbers in App. A.

Word meaning: T3 is much better (Tables 5, 6)

- **WS-353:** 0.104 → 0.577 on covered pairs ($\Delta\rho$ = +0.474, CI [0.36, 0.61]) and 0.048 → 0.452 on all 353; all-but-the-top adds another +0.105 (0.682).
- **Analogies:** 0.51% → 67.7% on covered questions (71.0% with ABTT) and 19.8% on all 19,544; T3 solves 3,848 questions T1 misses, T1 solves 0 that T3 misses.
- **Rare words too:** ρ = 0.577 on all covered pairs vs. 0.661 on frequent ones – BERT's knowledge comes from pre-training (≈3.3B words), not from EWT, so this win is expected.
- **Agrees with the literature:** our best WS-353 layer is layer 1 (ρ = 0.690, ≈6.8k contexts); Bommasani et al. (2020) report 0.683 at layer 1 for bert-base-uncased with 10k contexts.

Tree structure: T3 is worse

- **20 required sentences:** T3 18.4% vs. T1 20.5% (Δ =-2.1 pt, CI [−6.4, 3.2]) – *not significant*; 20 sentences cannot separate them.
- **660-sentence pool:** T3 17.9% vs. T1 24.8% (Δ =-6.9 pt, CI [−8.0, −5.8]) – significant, and barely above a random tree (17.1%).
- **Reference:** the T2 probe reaches 81.2% / 82.9% on the same sentences.
- **So the two evaluations disagree:** pooling from the syntax layer gives much better vectors by the standard benchmarks and worse ones by the structural check.

Is the syntax layer the right layer for static semantics? No (Fig. 2)

- **Best static layers are the lowest:** WS-353 peaks at layer 1 (0.690), analogies at layer 2 (80.3%); even layer 0 (essentially the input embedding table) gives ρ = 0.685 and 72.1%. This matches Bommasani et al. (2020).
- **Anisotropy explains the decline:** raw WS falls with depth while the mean cosine of random word pairs rises from 0.04 to 0.40 (layer 11); after all-but-the-top the WS curve is almost flat (0.64–0.71).
- **Analogies still decline after ABTT:** upper layers are more context-specific.
- **No layer helps trees:** static MST UUAS is 15–21% at every layer.

representation for the MST	tier	20 sentences [95% CI]	pool (660) [95% CI]	edge AUC
random tree (expected)	–	15.2	–	–
linear chain	–	43.2 [40.6, 46.4]	43.0 [42.4, 43.7]	0.75
T1 PPMI-SVD (spec)	req.	20.5 [16.3, 26.2]	24.8 [23.7, 26.0]	0.54
T1 PPMI-SVD (p0)	B	22.6 [17.8, 28.4]	24.5 [23.4, 25.6]	–
T3 BERT-static L8	req.	18.4 [13.6, 26.0]	17.9 [16.8, 18.9]	0.46
T3 BERT-static L8 + ABTT	B	19.7 [14.9, 26.2]	19.6 [18.6, 20.7]	0.44
concat [T1 ; T3] (L2-normalised)	C	19.2 [15.2, 25.1]	25.1 [24.0, 26.2]	–
T3 BERT-static L8 through probe B	C	29.5 [23.7, 38.2]	35.9 [34.3, 37.5]	–
contextual L8, cosine (no training)	B	35.9 [32.1, 39.6]	38.2 [37.3, 39.2]	0.70
T2 distance probe L8 k=128	ref.	81.2 [75.9, 85.9]	82.9 [81.9, 83.9]	0.96

Table 7: The diagnostic ladder on the same sentences. Edge AUC: within each pool sentence, the probability that a gold edge pair is closer than a non-edge pair (0.5 = no information), averaged over sentences. Tier: req. = required, B/C = our extensions.

	T1 PPMI-SVD	T3 layer 8	T3 + ABTT	random / gold
NN with same majority UPOS (top-10)	0.326	0.740	0.710	0.223
NN that co-occur within ±5 in train	0.557	0.162	0.156	0.067
hubness: skewness of N_{10}	2.10	1.45	0.57	–
function words among top-50 hubs	0.10	0.12	0.02	–
edge AUC (gold edge closer than non-edge)	0.54	0.46	0.44	0.50
MST: mean degree of function-word tokens	1.84	2.29	2.04	1.05 (gold)
MST: mean max. degree per sentence	3.75	5.46	3.62	4.53 (gold)
MST: edges joining same-UPOS words	0.21	0.25	0.29	0.13 (gold)

Table 8: Diagnostics (`diagnostics_task3.py`). NN statistics use the 2,134 types with train frequency ≥10 and the majority UPOS of each type; hubness uses the 3,919 types with frequency ≥5; MST statistics use the 660-sentence pool.

5.1 Why the evaluations disagree – measured, not assumed

1. **The tree lives in the context, and pooling averages it away** (Table 7).
   - The ladder adds one ingredient per rung: static T3 17.9% → the same vectors through the probe's subspace B 35.9% → per-token contextual vectors, plain cosine 38.2% → the trained probe 82.9%.
   - Averaging "bank" over all its sentences gives a good dictionary entry, but cannot know what "bank" attaches to in *this* sentence.
   - Even raw contextual vectors stay below the linear chain (43.0%): BERT's tree is in a learned low-rank subspace, not in its raw cosine geometry – exactly the structural probe's claim.
2. **"Similar" is not "connected" (paradigmatic vs. syntagmatic, Sahlgren 2006).**
   - **T3 finds substitutes:** 0.74 of its 10 nearest neighbours share the word's part of speech (T1 0.33; random 0.22), e.g. good → excellent, great (Table 9).
   - **T1 finds partners:** 0.56 of its neighbours co-occur with the word within ±5 (T3 0.16; random 0.07), e.g. said → spokesman, car → driving.
   - **Direct test, edge AUC:** inside a sentence, T3 closeness carries *no* information about attachment (0.46, slightly below chance); T1 a little (0.54); contextual vectors (0.70), word order (0.75) and the probe (0.96) are informative.
   - **Per relation:** T1 beats T3 on head–dependent pairs – nsubj (0.23 vs. 0.09), obj (0.31 vs. 0.15), amod (0.40 vs. 0.16); T3 wins only on conj (0.44 vs. 0.34), the relation that links two *similar* words ("cats and dogs").
   - **A first guess, corrected:** "gold edges join different categories" is *not* supported – gold edges are same-UPOS slightly *more* often than non-edges (0.13 vs. 0.09); the AUC result is what matters.
3. **Within-sentence hubs.**
   - Over the whole vocabulary T3 is *not* more hub-prone than T1 (skewness of the 10-NN in-degree N_{10} (Radovanović et al., 2010): 1.45 vs. 2.10).
   - Inside sentences T3's MSTs are star-shaped around frequent function words: function-word tokens get 2.29 edges on average (gold 1.05, T1 1.84); the largest degree per sentence is 5.46 (gold 4.53).
   - All-but-the-top reduces both (Table 8) and gives the significant +1.7 pt on the pool, but does not fix the

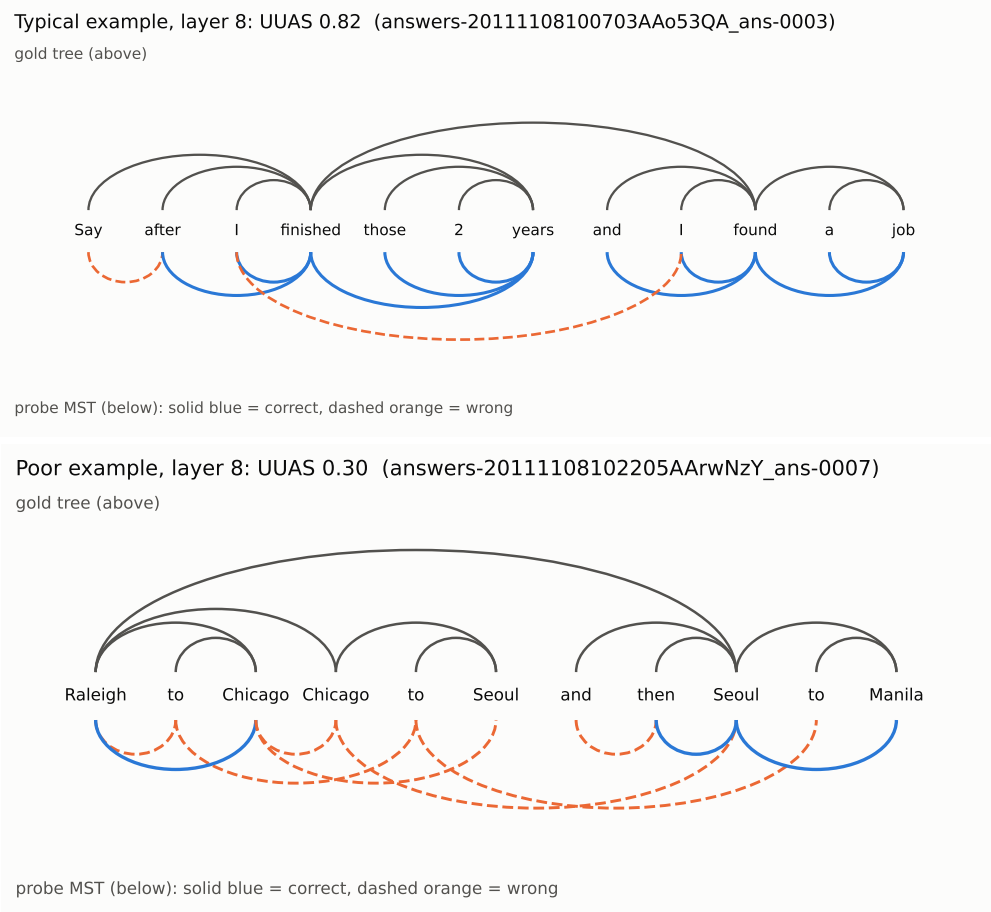


Figure 3: T2 probe (layer 8) on the median (top) and the worst (bottom) test sentence of 10–18 words. Gold tree above the words, probe MST below: blue solid = correct, orange dashed = wrong.

tree.

Combination variants

- **Concatenation [T1; T3]** hurts semantics (WS 0.397, analogy 22.2% – T1's noisy similarities dilute BERT's) and only matches T1 on trees (25.1%): the two spaces are not usefully complementary.
- **T3 through the probe's** B doubles its tree score (35.9%): the syntactic subspace carries some *type-level* information (which kind of word attaches to which), but most of the tree requires context.

6 Qualitative analysis

Task 2: probe errors (examples chosen by rank among test sentences with 10–18 words, not by hand)

- **Best** (UUAS 1.00): "I have eaten here several times and everytime the service is slower than slow." – every edge correct.
- **Median** (0.82): "Say after I finished those 2 years and I found a job.".
  - missed: Say-finished (discourse), finished-found (conj); spurious: Say-after, I-I;
  - the probe joins the two subjects "I" to each other instead of linking the clauses through their verbs, and misreads the discourse use of "Say".
- **Worst** (0.30): the verbless itinerary "Raleigh to Chicago, Chicago to Seoul and then Seoul to Manila.".
  - the probe links *repeated identical words* (to–to, Chicago–Chicago) and chains the cities left to right, whereas UD attaches each conjunct to the first;
  - identical word forms get very similar BERT vectors, so lexical identity leaks into the syntactic distance – the most frequent failure we saw, together with coordination and PP attachment.

word	T1 PPMI-SVD: 5 nearest neighbours	T3 BERT-static (layer 8): 5 nearest neighbours
good	nice, food, pretty, great, really	excellent, great, nice, decent, wonderful
bad	faith, big, too, wrong, something	terrible, horrible, poor, good, shitty
monday	friday, 0832, july, sunday, 6	tuesday, sunday, thursday, wednesday, saturday
dog	got, horse, cat, myself, leave	dogs, cat, puppy, animal, cats
car	home, guy, test, driving, drive	cars, vehicle, vehicles, suv, ferrari
money	save, enough, much, else, attention	cash, funds, dollars, cashout, bucks
said	spokesman, nun, penman, gregg, maviglio	says, stated, saying, told, say
happy	definitely, customer, guys, !!, safe	pleased, glad, unhappy, satisfied, delighted
computer	disc, plans, evidence, read, mohammed	laptop, computers, pc, machine, internet
two	screamed, paddled, expressway, kitchens, revolt	three, four, 2, five, six

Table 9: Nearest neighbours (cosine). T1 finds co-occurring partners (and noise from rare words); T3 finds substitutes.

section	question (a : b :: c : **gold**)	T3 answer	T1 answer
capital-common-countries	baghdad : iraq :: beijing : **china**	china ✓	jintao
capital-world	amman : jordan :: baghdad : **iraq**	iraq ✓	threats
city-in-state	houston : texas :: philadelphia : **pennsylvania**	pennsylvania ✓	franchise
family	boy : girl :: brother : **sister**	sister ✓	cousin
gram1-adjective-to-adverb	amazing : amazingly :: complete : **completely**	completely ✓	digest
gram2-opposite	acceptable : unacceptable :: aware : **unaware**	unaware ✓	socially
capital-world	beijing : china :: kingston : **jamaica**	jamaican	northwest
currency	brazil : real :: canada : **dollar**	actual	church
gram2-opposite	acceptable : unacceptable :: certain : **uncertain**	specific	socially
gram3-comparative	bad : worse :: cool : **cooler**	nicer	dried
gram4-superlative	bad : worst :: big : **biggest**	largest	chinese
gram6-nationality-adjective	australia : australian :: england : **english**	british	soper

Table 10: Analogy examples: top, questions T3 solves and T1 does not (one per section); bottom, T3 failures.

Task 3: neighbours, analogies and one sentence

- **Neighbours** (Table 9) show the paradigmatic/syntagmatic contrast directly; T1's lists for mid-frequency words (*two*, *computer*) are noise from rare co-occurrences.
- **Analogy successes** (Table 10): near-perfect on regular morphology (plural 96.6%) and frequent capitals (93.6%).
- **Analogy failures:** ambiguous or rare answers – currency (0/28; "real" occurs 53 times in EWT train, always as an adjective) and opposites (26.9%).
- **Near misses:** T3 picks a semantic neighbour over the morphological target (cooler → nicer, biggest → largest, english → british) – the substitute bias again. T1's wrong answers are unrelated co-occurring words (Baghdad:Iraq::Beijing → *jintao*).
- **A representative sentence** (Fig. 4; chosen as the one whose per-method scores are closest to each method's average): T1 and T3 each get 25%, raw contextual vectors 33%, the probe 83%; the T3 tree is a star around "a" and "of".

7 Discussion

What worked

- The structural probe reproduces on a different treebank, genre and annotation scheme: same peak layers, same rank saturation, within 0.3 pt UUAS.
- Pooled BERT vectors are strong lexical-semantic vectors even from 200k words of text, because the knowledge was learned in pre-training.
- One cheap post-processing step (all-but-the-top) adds a significant +0.1 ρ.
- Paired tests, the 660-sentence pool and bootstrap CIs separated real effects from noise (the 6–8 layer plateau; the non-significant 20-sentence T1/T3 difference).

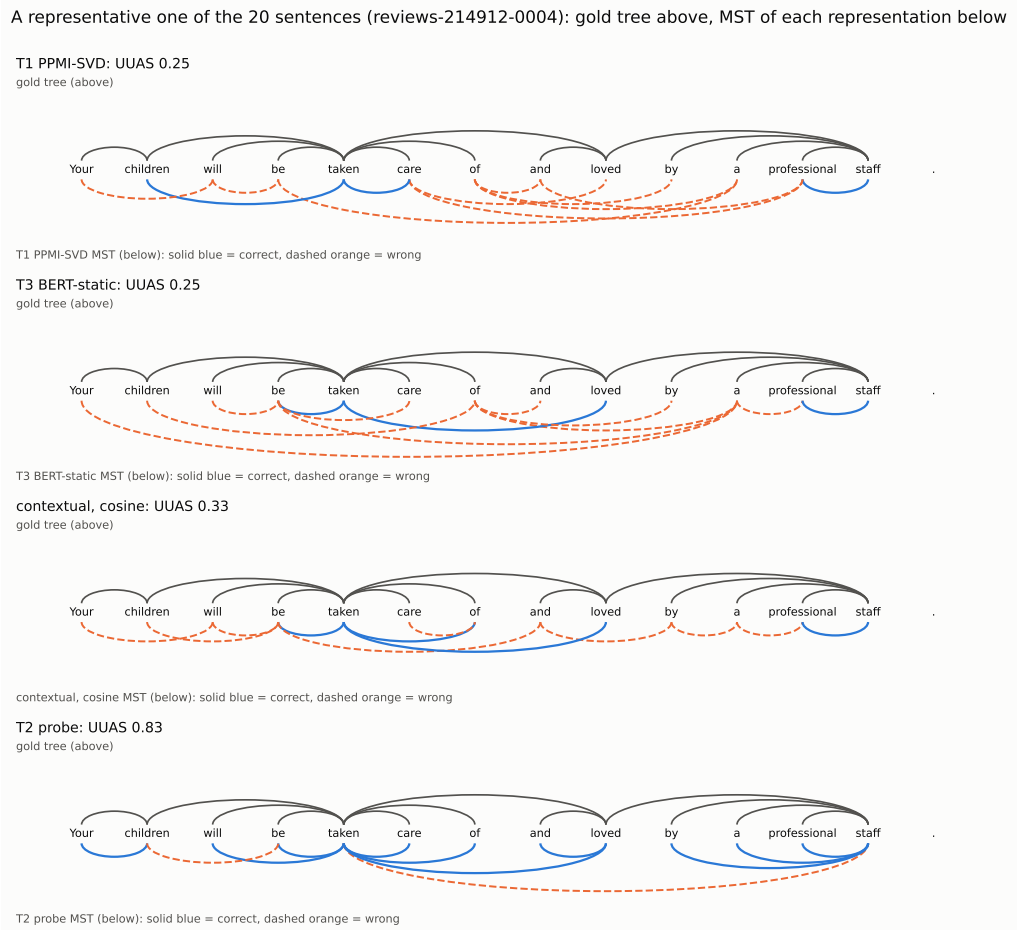


Figure 4: "Your children will be taken care of and loved by a professional staff." – gold tree (above) and the MST of each representation (below).

What did not work, and why

1. **PPMI-SVD on 200k words** cannot do analogies and only weakly captures similarity – data sparsity, confirmed by the frequency analysis and by the failure of every Levy et al. variant.
2. **Pooling the syntax layer does not give syntactic vectors.** The structural check rewards *syntagmatic* association between words of one sentence; averaging removes the token-specific information that makes BERT syntactic, leaving substitute-style similarity that is uninformative about attachment (edge AUC 0.46).
3. **The layer is mismatched:** the syntax layer (8) is not the best static-semantics layer (1–2).
4. **The spec's constant learning rate** does not let an L1-loss probe converge.

Do the two answers agree? No

- Semantic benchmarks measure paradigmatic similarity between word *types*; the tree check measures syntagmatic relations between *tokens*.
- A representation can – and here does – improve on one while getting worse on the other.

Limitations

- One treebank and one model.
- The 20-sentence check is under-powered (hence the pool), and the sample is biased towards in-vocabulary, 5–40-word sentences.
- BERT brings 3.3B words of external knowledge, so T1 vs. T3 is not a same-data comparison.
- The causes of the DSpr/NSpr gap are plausible but not isolated.

8 Reproducibility, seeds and deviations

- **Seeds:** `SEED = 42` (`code/common/config.py`) seeds Python, NumPy and PyTorch (`PYTHONHASHSEED` set). It drives the 20-sentence sample (`numpy.random.default_rng(42)`), the random-tree baseline, all bootstraps (1000 resamples), the SVD (`random_state=42`) and the anisotropy sampling. Probe (ℓ, k, type) is initialised with seed 42 + 1000ℓ + k (+500 for depth); batch order uses seed 42 + ℓ.
- **Environment:** Python 3.12, PyTorch 2.15, Transformers 5.22, Datasets 5.1 (pinned in `requirements.txt`); 2×NVIDIA L40S (shared).
- **Runtime:** T1 seconds; BERT features 7 min; 78 probes 49+54 min on two GPUs; T3 <5 min.
- **Pretrained checkpoint:** `google-bert/bert-base-uncased`, Hugging Face Hub, revision `86b5e09`.
- **Code:** `code/task1`, `task2`, `task3` (plus `common/` for data loading and the shared evaluation); `run_all.py` and a Colab notebook run everything end-to-end (README).

Deviation / interpretation	Justification
Window "size 5" = 5 words each side	Standard reading (Levy et al., 2015); counts never cross sentence boundaries.
Lowercasing (T1, T3, benchmarks)	The model is uncased; T1 and T3 then share one vocabulary, so their coverage is identical.
WS-353 reported on all 353 pairs *and* on the 248 covered	"All 353" is the spec-literal number (OOV pairs get the median similarity, a neutral rank; a 0 would be non-neutral for anisotropic BERT); "covered" is the informative comparison. Analogy: OOV questions count wrong in "all".
20-sentence sample: 5–40 words, all in vocabulary; punctuation ignored	A static model has no vector for an OOV word; punctuation is excluded exactly as in the probe paper. The bias towards simpler sentences is acknowledged.
LR ×0.1 on dev plateau + best-dev checkpoint (T2)	The paper's schedule; the spec-literal control (§4.2) keeps ℓ^* but does not converge.
Layer 0 probed in addition to 1–12	Non-contextual baseline (cf. the paper's PROJ0).
ℓ^* chosen on dev, not test	Avoids test-set selection; the 6–8 plateau makes the choice robust.
Extra analyses (pool, CIs, paired tests, sweep, ABTT, ladder, diagnostics)	Additions only; every required number is reported as specified.

Table 11: Deviations from the task specification and their justification.

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A Additional tables

	WS-353 ρ (cov.)		Analogy (cov., %)		MST UUAS on the pool (%)				
ℓ	raw	ABTT	raw	ABTT	static raw	static ABTT	contextual cos.	T2 probe	anisotropy
0	0.685	0.703	72.1	69.4	17.0	16.4	23.2	63.8	0.04
1	0.690	0.686	75.9	74.0	16.7	17.0	24.6	71.1	0.09
2	0.675	0.693	80.3	79.7	15.4	17.3	24.0	73.3	0.13
3	0.658	0.682	78.0	78.3	15.6	16.9	23.9	77.9	0.16
4	0.602	0.644	77.1	77.6	15.9	17.2	26.7	82.3	0.21
5	0.587	0.645	74.1	75.4	15.3	17.3	28.3	83.1	0.25
6	0.578	0.647	72.3	73.7	15.5	17.6	30.7	83.8	0.27
7	0.577	0.666	70.4	73.0	16.4	18.2	34.1	83.6	0.28
8	0.577	0.682	67.7	71.0	17.9	19.6	38.2	82.9	0.29
9	0.592	0.713	66.1	69.2	19.2	20.1	38.8	79.4	0.29
10	0.552	0.711	62.9	67.5	19.8	20.5	37.4	75.0	0.32
11	0.488	0.706	62.1	68.2	19.0	20.8	36.5	72.3	0.40
12	0.528	0.691	63.7	69.0	18.9	20.2	34.4	70.5	0.33
T1	0.104	0.153	0.5	0.4	24.8	24.5	–	–	0.12

Table 12: Layer sweep for pooled static vectors (covered items; pool = 660 sentences), with contextual references.

B Additional figures

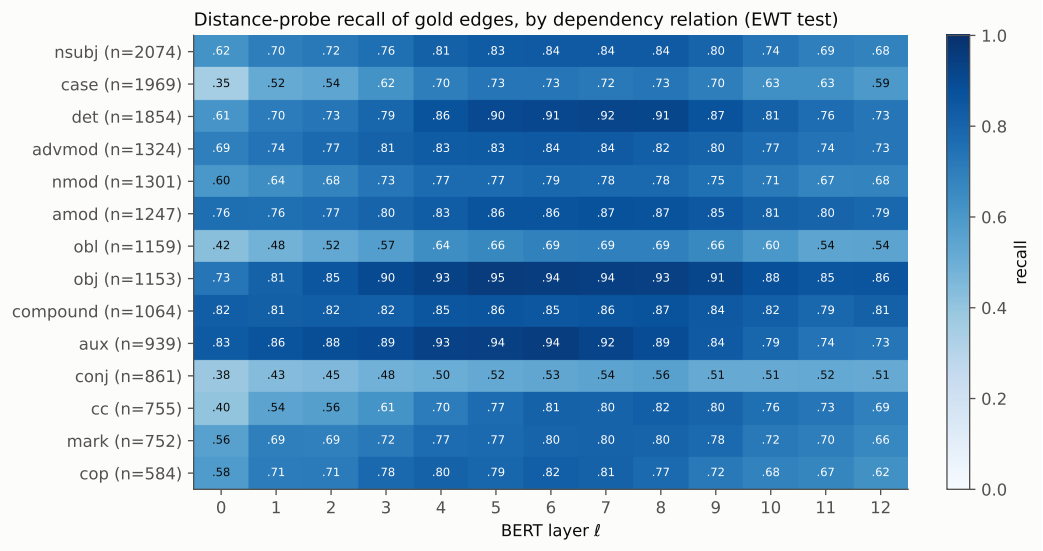


Figure 5: T2 distance-probe recall per dependency relation and BERT layer (EWT test, dev-best rank per layer).

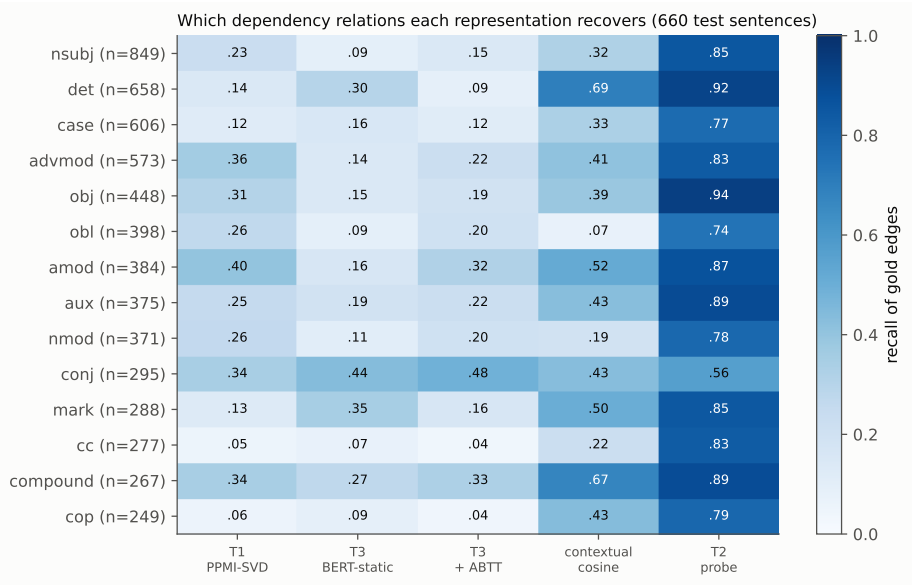


Figure 6: Recall of gold edges per relation for each representation on the 660-sentence pool.

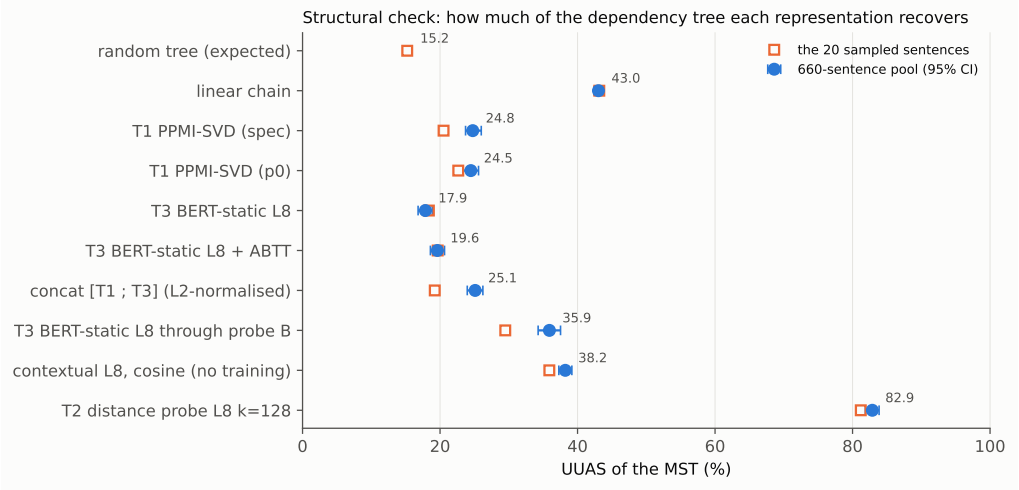


Figure 7: The diagnostic ladder (Table 7): pool UUAS with 95% CI (filled) and the 20 sentences (open squares).

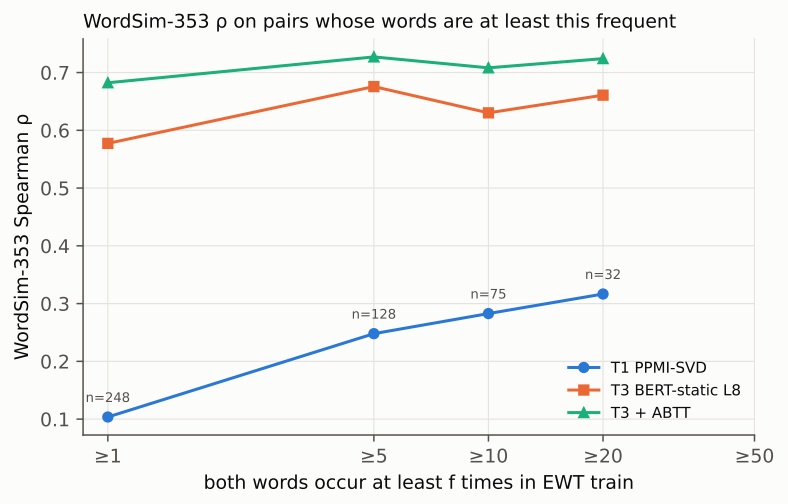


Figure 8: WS-353 ρ on pairs whose two words occur at least f times in EWT train (n = number of pairs).

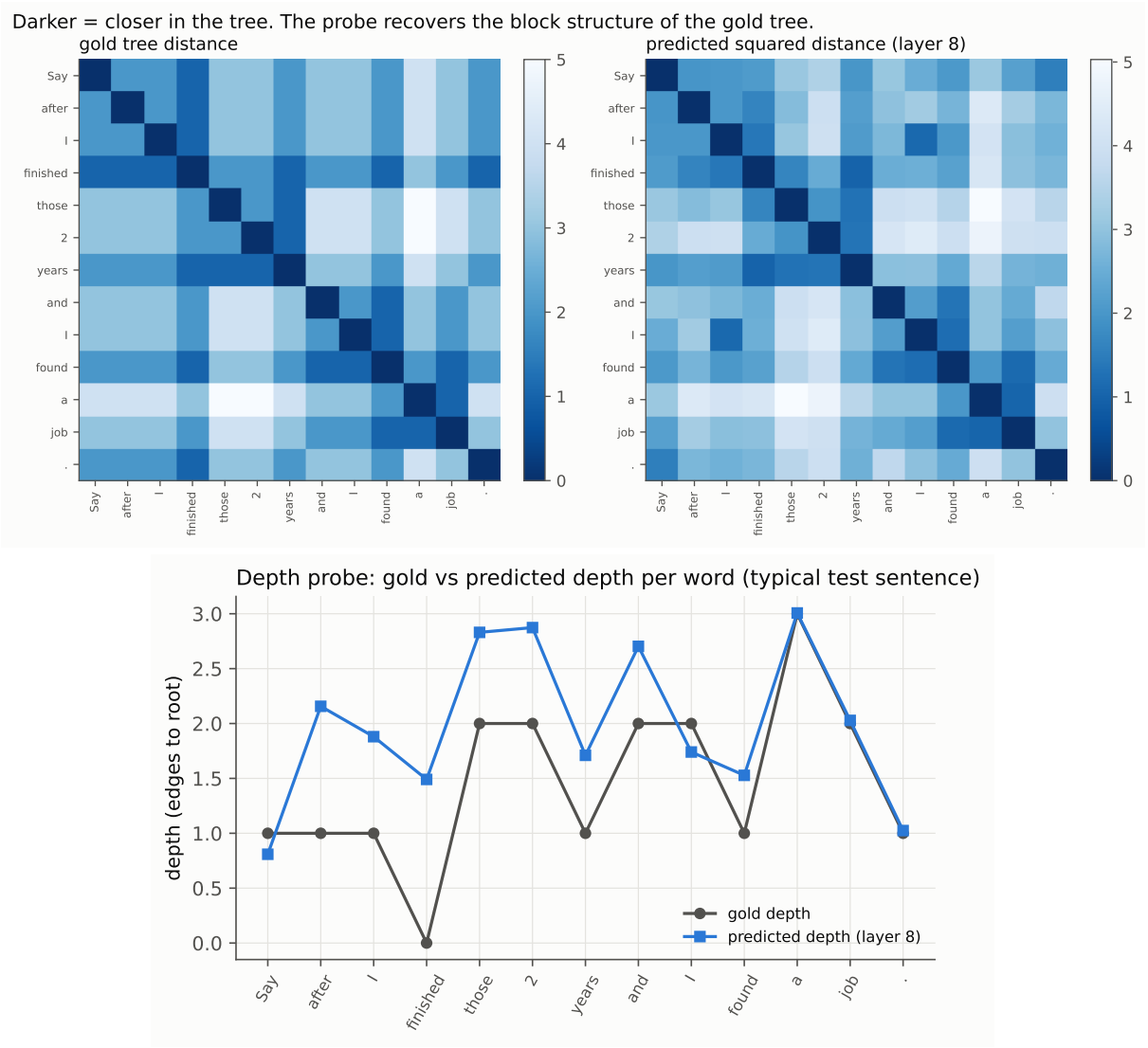


Figure 9: T2 on the median sentence: gold tree distances vs. predicted squared distances (top), gold vs. predicted depth (bottom).